

Traffic Sign Classification

Deep Feature Manifolds & Hybrid SVM Optimization

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Final Project Report — 2026

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Introduction

Project Objectives

- **Goal:** Build a robust traffic sign recognition pipeline for autonomous vehicles.
- **Key Approach:** Bridge the gap between Deep Learning and Support Vector Machines.
- **Innovation:** Investigation of Low-Dimensional (3D) Manifold Classification.

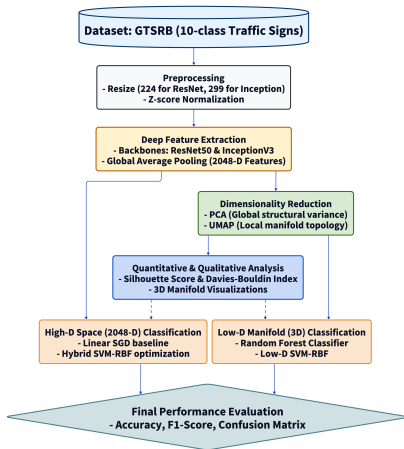
Dataset: GTSRB (10-class Subset)



- Over 10,000 images across 10 distinct traffic sign categories.
- Significant scale and illumination variations.

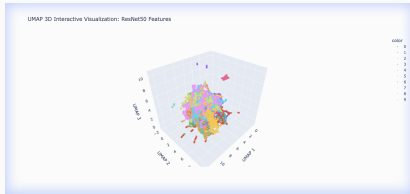
Methodology

Pipeline Overview

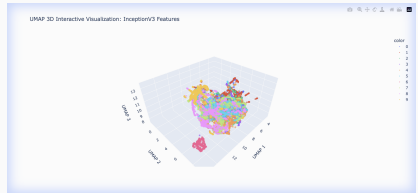


1. **Data & Preprocessing:** Stratified 10-class selection; Z-score normalization.
2. **Feature Extraction:** Deep CNN backbones as 2048-D semantic encoders.
3. **Geometric Analysis:** Contrast of Global (PCA) vs. Local (UMAP) topology.
4. **Dual-Track ML:** Evaluation on High-D space vs. Compressed 3D manifolds.

3D Manifold Visualizations (UMAP)



ResNet50 Manifold



InceptionV3 Manifold

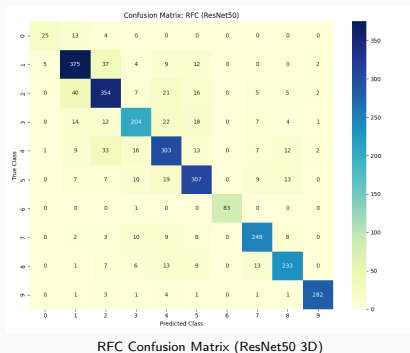
UMAP preserves local structures, forming dense semantic clusters.

Results & Analysis

Classification in 3D Space

- **Insight:** Linear models fail in 3D (<35%).
- **Success: Random Forest** dominates by recursive partitioning.

3D Classifier	Acc. (R50)
SGDC (Linear)	28.94%
SVM-RBF	51.74%
Random Forest	82.28%

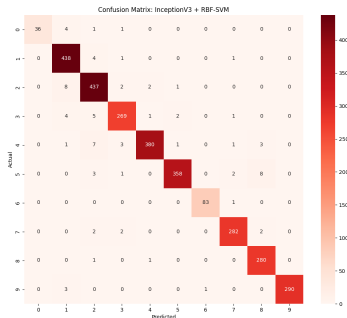


High-D Classification Results

Model	R50	IncV3
SGD (Lin)	91.2%	88.5%
RBF (Hyb)	95.9%	97.2%

Validation Acc.

RBF kernels effectively handle the density of deep manifolds, significantly outperforming linear baselines.



Peak Model CM (InceptionV3 + RBF)

Conclusion

Summary

- Hybrid architectures improve flexibility and performance.
- **RBF kernels** are essential for 2048-D features (achieving up to **97.24%** accuracy).
- 3D projections retain **82% accuracy**, proving UMAP's density preservation.

GitHub Repository:

<https://github.com/toando0212/final-ml2>

Questions?